

TIRATH BHATHAWALA

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EDUCATION

Universiteit van Amsterdam Amsterdam, Netherlands
Master of Science in Artificial Intelligence Sept 2026 – Jun 2028

SVKM's Dwarkadas J. Sanghvi College of Engineering Mumbai, India
B.Tech in Computer Engineering with Honors in Intelligent Computing (CGPA: 9.32) Nov 2022 – Jun 2026

RESEARCH PUBLICATIONS

Dynamic Optimization of Peer Review Length Using Information Density Analysis Published
Scientometrics (Springer) — [Paper](#) — [Code](#) — [Demo](#) September 2025

FRAC-MAS: A Safe and Explainable Multi-Agent System for Fracture Diagnosis Accepted
The 2nd Agentic AI for Medicine Workshop at MICCAI 2026 — [Website](#) — [Code](#) — [Demo](#) September 2026

DigniFy: Multi-Modal Multilingual Online Hate Speech Detection Accepted
IEEE CONECCT 2026 — [Code](#) — [Demo](#) June 2026

AntiBug: Runtime-Aware Multimodal Agentic Program Repair for Production Applications Under Review
Automated Software Engineering (Special Issue) — [Website](#) — [Code](#) April 2026

EXPERIENCE

BrandContext.ai Mar 2025 - Aug 2025
AI Intern Hybrid, Mumbai, India

- Fine-tuned an **LLM-driven automation framework (browser-use)** that independently performs user-specified queries on **websites (Looker Studio, Hootsuite, etc)** or Google Sheets by generating and executing custom **Playwright-based actions**.
- Developed an **AI-agentic system** powered by **dynamic graphs** and **real-time factory operation databases** to support **supervisor workflows**, **detect anomalies**, and dynamically incorporate **worker-input special conditions**.
- Evaluated and productionized computer-vision pipelines for retail supermarkets (**queue, theft, etc**) and fine-tuned **YOLOv11** multi-class detectors (**logo, furniture, mobile, security**) for **Xiaomi (0.87 mAP@50 across over 500 stores in India)**.
- Built legally-compliant **speaker-diarization** pipelines for sales-customer calls using open-source (**SpeechBrain**) and commercial (**ElevenLabs**) stacks to enable accurate speaker labeling and downstream analytics.

Indian Institute of Technology, Patna Oct 2024 - May 2025
Research Intern Remote, Patna, India

- Developed a heuristic framework based on **Cognitive Load and Discourse Analysis** to optimize peer review lengths using information density, content relevance, argument strength, and readability metrics. **Published in *Scientometrics***.
- Designed **MetaSearch**, a **search-augmented, reasoning-based AI Agent** for **consensus resolution in peer review**, combining **disagreement detection** and **fact-grounded synthesis** to automate and enhance meta-review decision-making.
- Developed a novel LLM-argumentation system for peer review, evaluating 900+ ICLR reviews (**PeerRead Dataset**) using leading LLMs (**Llama 3.1, Mixtral, Gemini2, GPT-4o**), achieving high inter-annotator agreement (**Cohen's Kappa = 0.834**).
- Engineered **BiasChain**, a **modular multi-agent LLM framework** for justified peer review **bias detection**, by orchestrating specialized agents for **sentiment coherence**, **internal consistency**, and **inter-review alignment**.

Nvelop Technologies Oy Aug 2024 - Feb 2025
AI Intern Remote, Helsinki, Finland

- Authored a whitepaper implementing an evaluation pipeline for **LLM-based requirement generation**, benchmarking **DeepSeek R1, GPT-4o, Gemini 2.0, LLaMA 3.2** across diverse **NLP metrics** including BLEU, Levenshtein and Jaccard Similarity.
- Devised automated workflows in **Azure AI Promptflow** for Retrieval Augmented Generation (RAG) tasks, for 50+ RFPs.
- Optimized chunking strategies and top-p sampling parameters, **reducing document retrieval perplexity by 15%**.
- Authored a **comprehensive 70-page documentation** detailing the mapping of AI APIs (**Insomnia**) to the frontend (**NextJS**).

UIUC+ Summer Research Program Jun 2024 - Jul 2024
Research Assistant Remote, Urbana-Champaign, USA

- Investigated challenges in **C to Rust transpilation**, summarizing findings from 5+ research papers (**VERT, Lost in Transpilation, etc.**) and evaluating the results of existing transpilers and CodeLLMs.
- Analyzed LLM-generated C and Rust code, identifying and categorizing bugs, improving **code quality evaluation frameworks**.

PavePilot AI

Python Development Intern

Jun 2024 - Jul 2024

Remote, USA

- Established an AI-driven marketing content pipeline by evaluating multiple image models (**BRIA, Stable Diffusion, Midjourney**) and implementing image processing techniques with **OpenCV**.
- Created platform-specific content (**LinkedIn, Instagram, Twitter**), designing customized AI-generated prompts and style guides for **brand-aligned product advertisements** using GenAI tools.

EXTRA CO-CURRICULAR

DJSCE ACM | *Research Head*

- Established personalized research mentoring program for **15+ students**, successfully placing **5 mentees in competitive research internships** through targeted guidance in domain-specific learning and academic paper writing.
- Designed tasks in **python, research methodology, and technical writing** to enhance mentees' skills in structured research.

PROJECTS

AntiBug: Runtime-Aware Multimodal Agentic Program Repair for Production Applications | *LangGraph, Next.js*

- Developed **AntiBug**, a runtime-aware automated program repair system for **React/Next.js**, with a client-side telemetry SDK capturing source-map-resolved stack traces, user-event sequences, and DOM screenshots with cursor context.
- Implemented a **12-agent LangGraph pipeline** with conditional triage, parallel specialist analysis, **root-cause hypothesis generation and validation**, targeted patch synthesis, and a self-healing validation loop.
- Evaluated on **69 real-world bugs** from Cal.com and Appsmith, fixing **32/69 issues vs 9/69 for GPT-4o**; improved mean fix and root-cause scores by **0.74 and 1.04**; additionally solved **20/40 relevant SWE-Bench Multimodal issues**.
- Achieved **72.5% preference** across **549 blind A/B evaluations** by 8 developers, beating baseline confidence (**4.26 vs 3.14**).

FRAC-MAS: A Safe and Explainable Multi-Agent System for Fracture Diagnosis | *PyTorch, LangGraph, Grad-CAM*

- Developed a **multi-agent clinical decision support system** for fracture diagnosis, combining a **4-model vision ensemble** (MaxViT, YOLOv26m-cls, RAD-DINO, HyperColumn-CBAM DenseNet-169) with **Grad-CAM explainability**.
- Integrated **conformal prediction** for statistically grounded differential diagnoses and a blind-first **Critic Agent** for escalation, achieving **92.0% empirical coverage** and **reducing escalations to 13.4% with 93.8% accuracy** on confirmed cases.
- Achieved a **96.4% fracture detection rate** on an external Roboflow dataset and demonstrated strong human evaluation, with clinician technical-accuracy ratings of **4.14/5** and a **55.4% top preference rate** over Llama, MedGemma, and Gemini.

DigniFy: Multi-Modal Multilingual Hate Speech Detection | *HuggingFace, Tensorflow, Render, FastAPI*

- DigniFy is an AI-powered **mobile application** for **multimodal, multilingual** hate speech detection.
- Fine-tuned **DistilRoBERTa** for English and Hinglish text, implemented **OCR-based text extraction** for images, designed an **audio processing pipeline** with STT-TTS, and integrated all modalities into a unified **video processing pipeline**.
- Engineered an **AI Agent** integrating all modality-specific models as tools, enabling detection through interaction with four LLMs (**DeepSeek R1, GPT-4o, Gemini 2.0, LLaMA 4.0**), and conducted comparative evaluation across model outputs.

TECHNICAL SKILLS

Relevant Coursework: AI, ML, DL, Data Structures & Algorithms, Data Warehousing & Mining, Advanced DBMS, OS

Programming Languages: Python, C, C++, Java, HTML/CSS, JavaScript, SQL, React, Rust, $\text{\LaTeX}$

Tools: Linux, Git, Kaggle, Google Colab, Docker, Azure AI, Overleaf

Frameworks/Technologies: Tensorflow, Keras, PyTorch, OpenCV, Transformers, NLTK, FastAPI, HuggingFace, scikit-learn

Course Certifications: CS50P by Harvard University (Online), The Odin Project

Coding Profiles: [CodeChef](#) [LeetCode](#) [HackerRank](#)

ACHIEVEMENTS

All India Rank 499 in GATE DA (Data Science and Artificial Intelligence) examination from over 60,000 students

Winner in CodeBounty 2024, a college competitive programming contest with 100+ registrations

Top 100 in the Rajasthan Police Hackathon 2024 (ML Lead) out of 1000+ teams

Top 200 in the Gujarat G20 Summit Hackathon 2023 (ML Lead) out of 2500+ teams

Rank 292 out of 70,000+ registrations in the **Amazon ML Challenge 2024**

Selected as a **mentee** in the [sktime](#) open-source program under [Dr. Franz Kiraly](#)